School of Computing Science & Engineering

(SCOPE)

AWS Practical Record File

AWS Cloud Practitioner (CSE3015)

Slot:C21+F11+F12

Class No.: BL2025260100719

Submitted By:

Name: Digvijay Singh

Reg. no.: 23BCE11015

Submitted to: Velmurugan L

AWS Cloud Practitioner

CSE3015

Hands-on Lab Manual

Preface

This manual consists of a series of laboratory experiments to accompany the syllabus of B. Tech. *Computing Science & Engineering,* **VIT Bhopal University**.

The experiments are divided among following headings:

- **Exercise (Objective)**
- **Prerequisites**
- **Objectives**
- **Underlying Concept / Theory (along with Figure if applicable)**
- **Problem Statement**
- **Algorithm**
- **Coding(if applicable)**
- **Output**
- **Remarks**
- **Conclusion**
- **Challenges**

The appropriate headings are chosen to be placed in experiments depending on type of experiment.

Indicative List of Experiments

Exp. No.	Title	Date Conducted	Remark
1	Launch an EC2 instance using the AWS Management Console by Select appropriate instance types, AMIs, and configurations. Connect to the EC2 instance using SSH or Remote Desktop.		
2	Create an S3 bucket and Configure bucket policies, versioning, and encryption. Upload and download objects to/from the S3 bucket.		
3	Create a Lambda function using Python or Node.js and Configure event triggers and test the Lambda function.		
4	Register a domain name using Amazon Route 53 and Upload HTML, CSS, and JavaScript files to the S3 bucket to host the website.		
5	Configuring a Virtual Private Cloud (VPC) in AWS, including setting up subnets, route tables, internet gateways, NAT gateway and security groups.		
6	RESTful API using Amazon API Gateway using AWS Lambda.		
7	Launch an RDS instance by Selecting a database engine (e.g., MySQL, PostgreSQL, SQL Server) and Configure database settings, including storage, security, and backups.		
8	Create an Identity and Access Management (IAM) in AWS, including creating users, groups, roles, and policies, and managing access permissions to AWS resources.		
9	Creating and managing AMIs and snapshots in AWS and Attach EBS on a running instance.		
10	Create CloudWatch alarms to monitor metrics EC2 instances, RDS databases, and ELB load balancers and set up dashboards to visualize performance metrics.		

Hands-on Lab Manual (Experiment 1)

Launch an EC2 instance using the AWS Management Console

Exercise 1

Launch an EC2 instance using the AWS Management Console by Select appropriate instance types, AMIs, and configurations. Connect to the EC2 instance using SSH or Remote Desktop.

Prerequisites:

You should already be familiar with:

- AWS Management Console
- Basic understanding of cloud computing and virtualization.
- For Windows instance: Remote Desktop Protocol (RDP) client.
- A key pair (can be created during the instance setup).

Objectives:

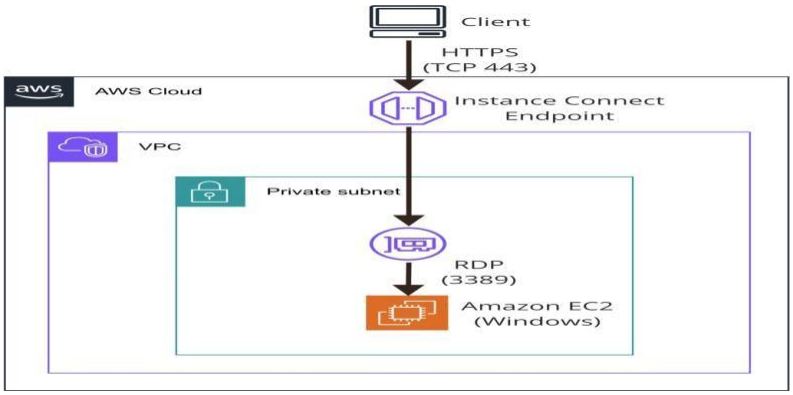
- Understand EC2 and its configuration options.
- Launch an EC2 instance with appropriate AMI and instance type.
- Configure security groups and key pairs.
- Successfully connect to the instance using RDP.

Underlying Concept / Theory:

Amazon EC2 (Elastic Compute Cloud) is a web service that provides resizable compute capacity in the cloud. It allows users to run virtual servers (instances) and manage them via the AWS Management Console or programmatically.

Key concepts:

- AMI (Amazon Machine Image): Pre-configured template with OS and software.
- Instance Type: Defines hardware specs like CPU, memory.
- Key Pair: Used for secure SSH or RDP login.
- Security Group: Acts as a virtual firewall.

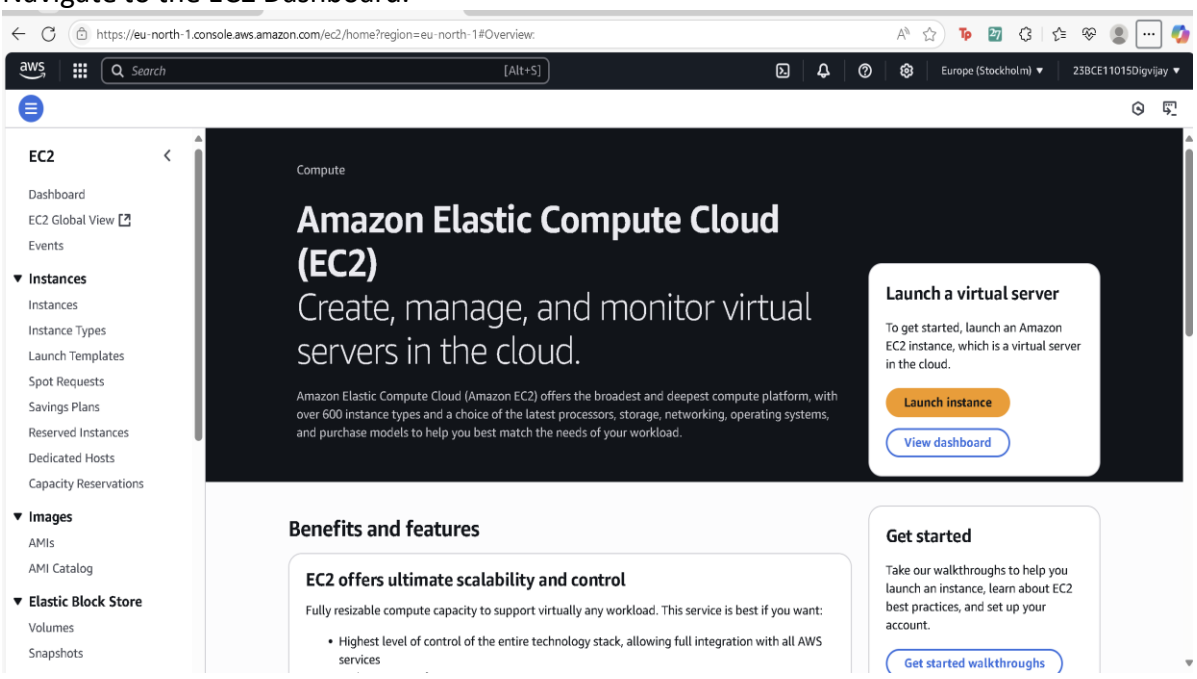


Problem Statement:

Launch and configure an EC2 instance from the AWS Management Console using a specified AMI and instance type. Then, connect to the instance using SSH (for Linux) or RDP (for Windows).

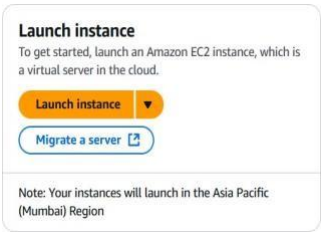
Procedure:

1. Login to AWS Console.
2. Navigate to the EC2 Dashboard.

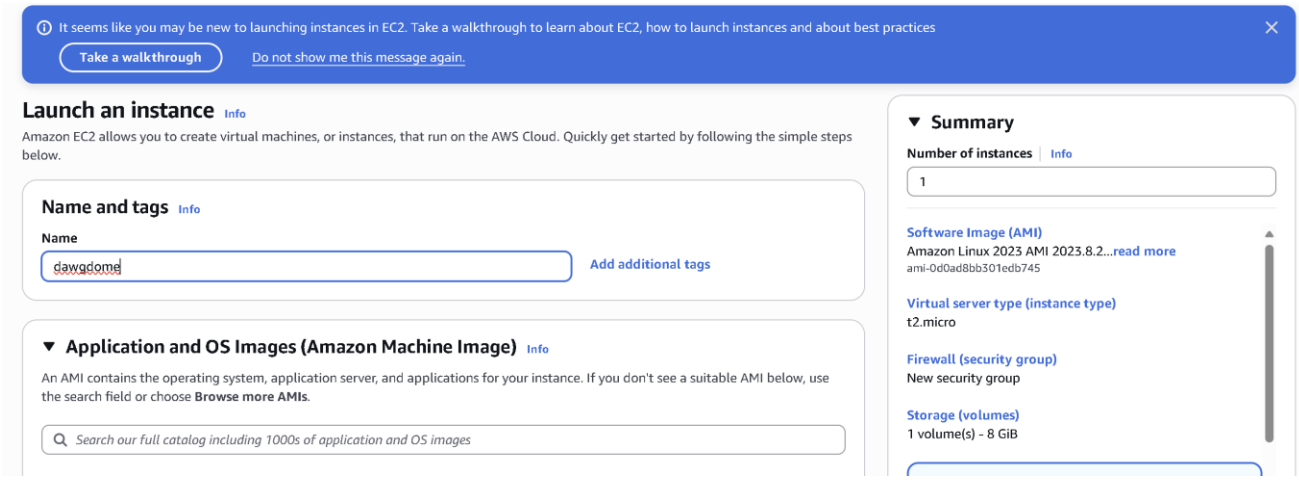


3. Select Region from the right top corner.

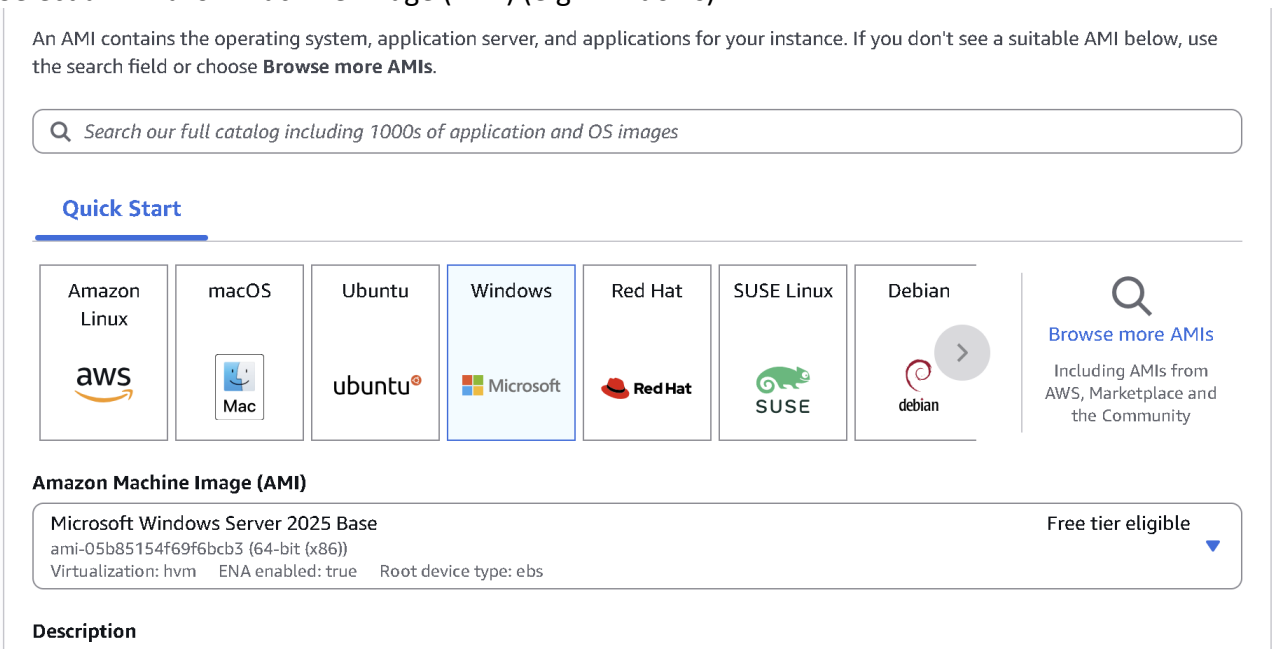
4. Click “Launch Instance”.



5. Enter a name of EC2 Instance.

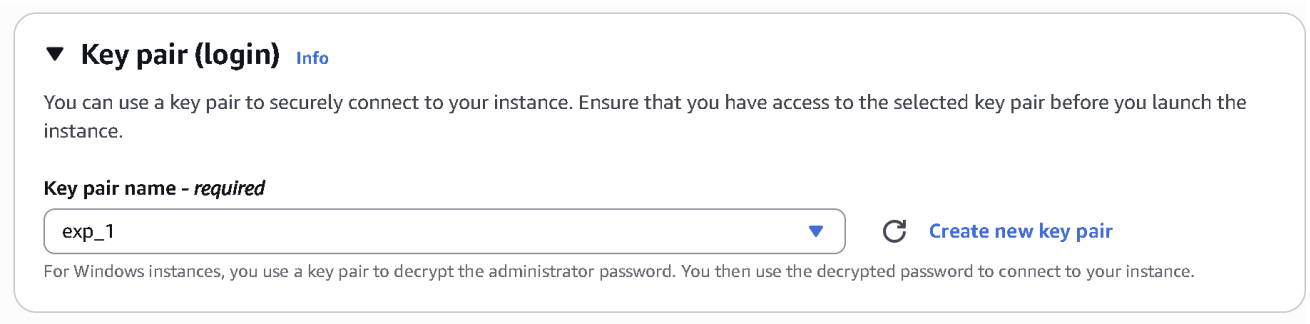


6. Select an Amazon Machine Image (AMI) (e.g. Windows).

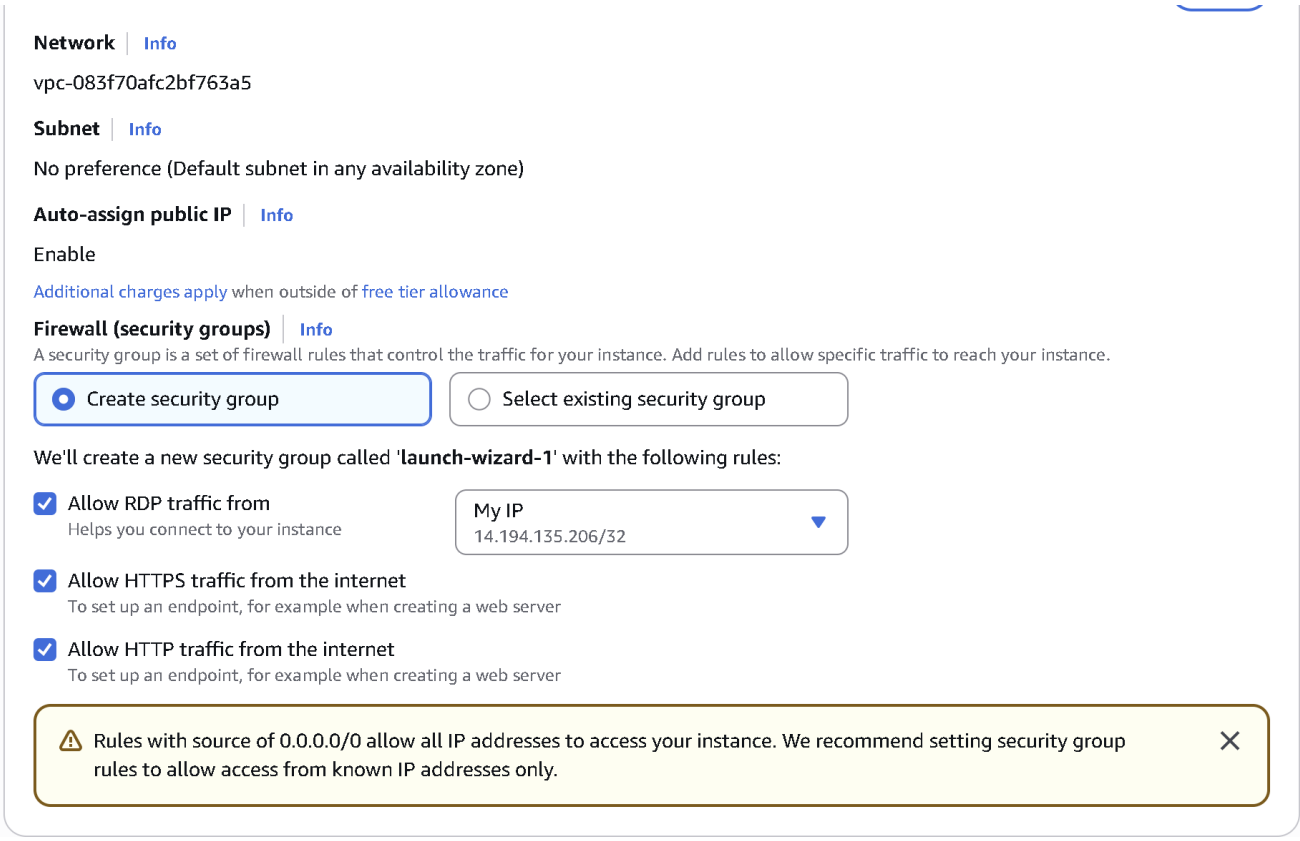


7. Select an Instance Type (e.g., t3.micro).

8. Create and select a Key Pair for secure connection.

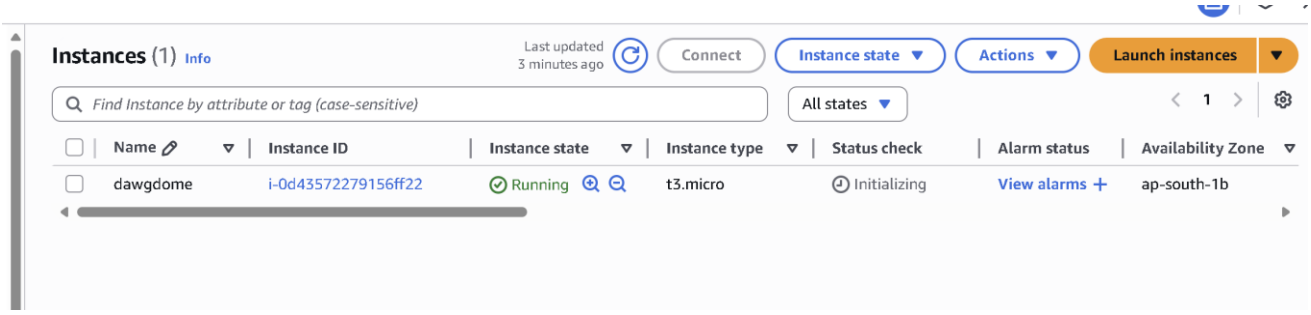


9. Configure Network Settings and allow ports.

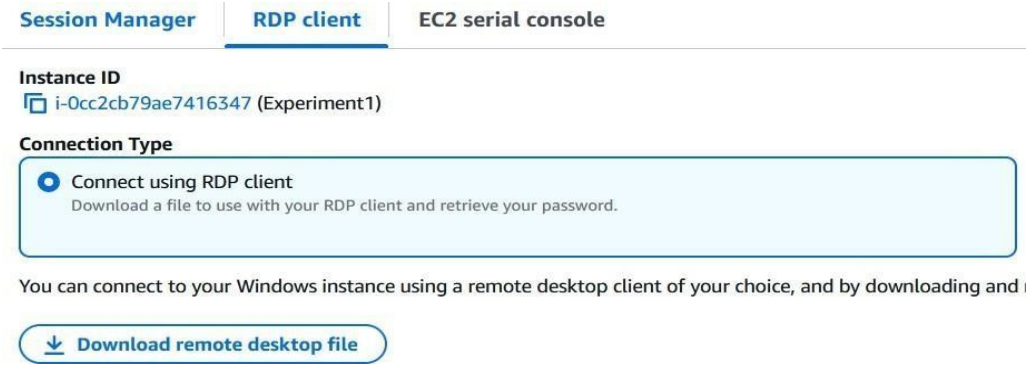


10. Review configuration and click “Launch Instance”.

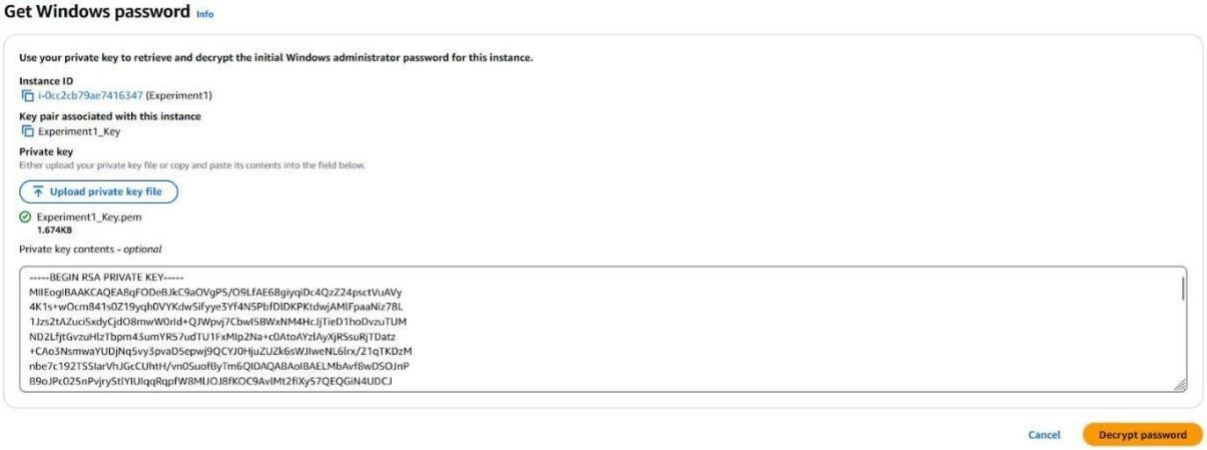
11. Wait for the instance status to show “Running”.



12. Select the Experiment1 checkbox and then connect it.
13. Goto RDP Client then download remote desktop file.



14. Click get password and upload Experiment1_Key and decrypt the password.

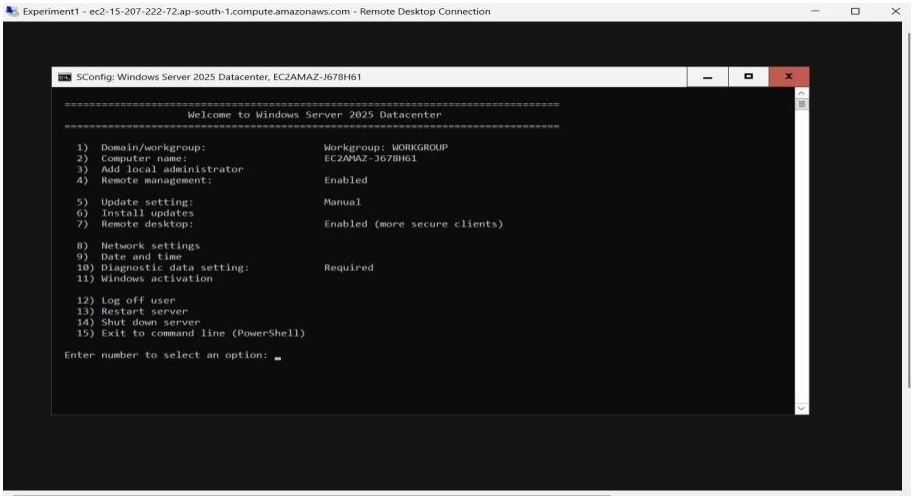


15. Run downloaded remote desktop file and enter the decrypted password.

16. Connect to the instance.

Output:

Successfully running an EC2 instance.



Remarks:

- Ensure the key pair is downloaded and securely stored.
- If the instance does not connect, verify security group settings (inbound rules).
- The free tier supports t2.micro instances for up to 750 hours/month.
- Don’t forget to terminate unused instances to avoid charges.

Conclusion:

The experiment demonstrates how to deploy and connect to a virtual server using Amazon EC2 RDP Client. This provides foundational experience in cloud infrastructure management and serves as a stepping stone toward more complex architectures and deployments using AWS.

Hands-on Lab Manual (Experiment 2)

S3 BucketExercise 2

To demonstrate the concept of creating and managing an Amazon S3 Bucket with policies, versioning, and encryption.

Prerequisites:
You should already be familiar with:

- Basic AWS services, especially Amazon S3.

- Using the AWS Management Console or AWS CLI.

- Basic concepts of cloud storage and permissions.

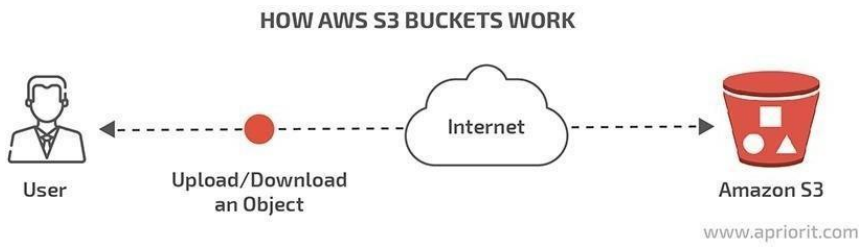
Objectives:
After completion of this lab student shall be able to:

- Create an S3 bucket using AWS Console.

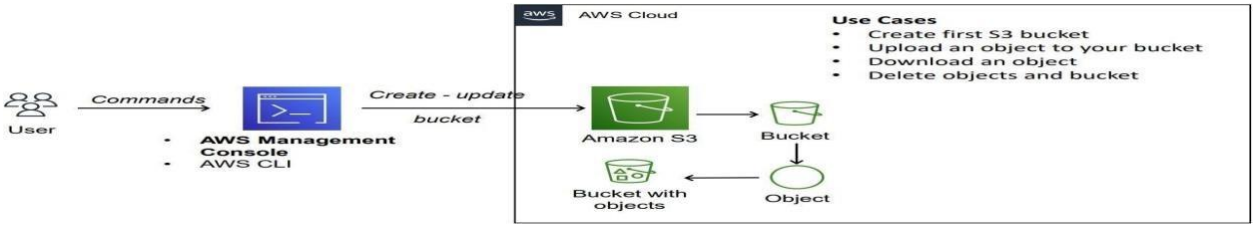
- Configure bucket policies for access control.

- Enable versioning and encryption for data protection.

- Upload and download objects to/from the S3 bucket.



Underlying Concept / Theory:
Amazon Simple Storage Service (S3) is a scalable object storage service provided by AWS. It allows you to store and retrieve any amount of data from anywhere on the web.



- Bucket: A container for storing objects (files).
- Bucket Policy: A JSON-based access policy attached to a bucket to define who can access it and how.
- Versioning: Allows you to preserve, retrieve, and restore every version of every object stored in a bucket.
- Encryption: Protects data by encrypting it at rest using server-side encryption (SSE-S3 or SSE-KMS).

Problem Statement:
Create an S3 bucket and configure it with appropriate bucket policies, versioning, and encryption. Upload and download objects to/from the S3 bucket.

Algorithm:

1. Create an S3 Bucket:
   - Go to AWS Console → S3 → Create Bucket.
   - Provide a unique bucket name and region.
2. Enable Versioning:
   - Go to “Properties” → “Bucket Versioning” → Enable.
3. Configure Bucket Policy.
4. Enable Encryption:
   - Under “Properties” → “Default encryption” → Choose SSE-S3 or SSE-KMS.
5. Upload Objects:
   - Go to the “Objects” tab → Upload → Select files from your computer.
6. Download Objects:
   - Select the uploaded file → Actions → Download.

Implementation Steps:

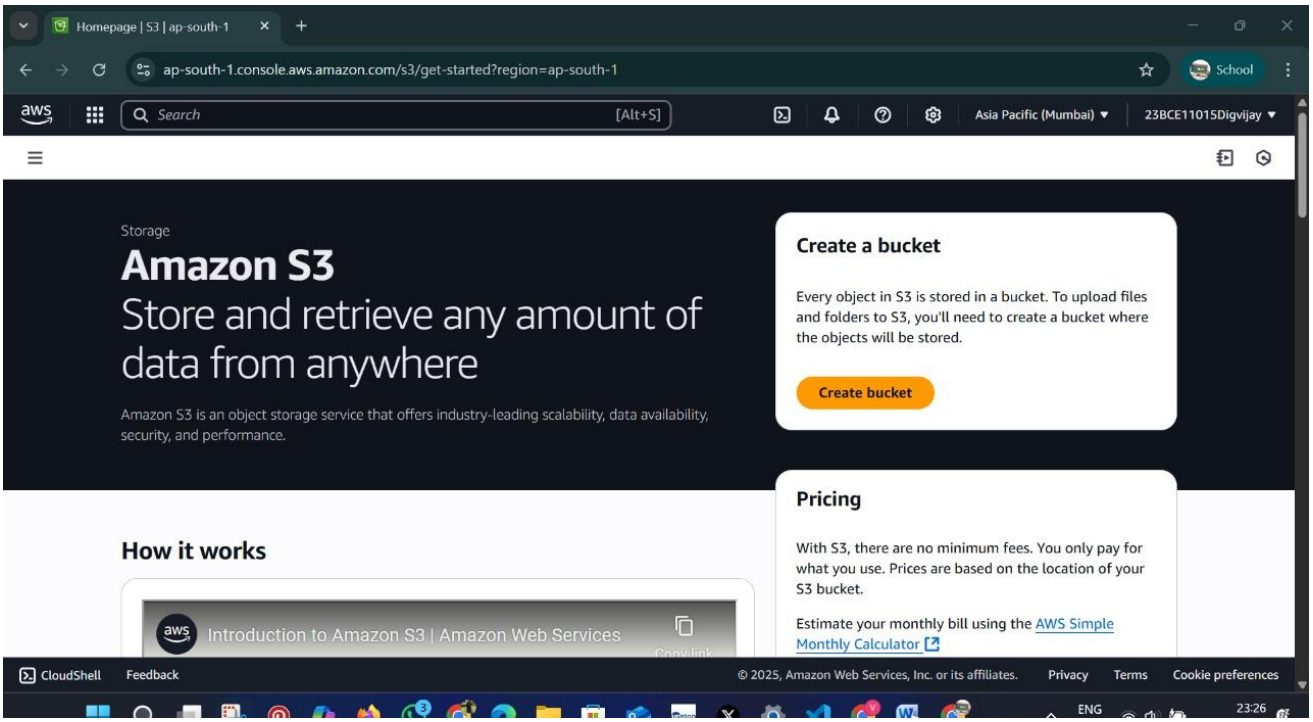


Fig 1: Create Bucket

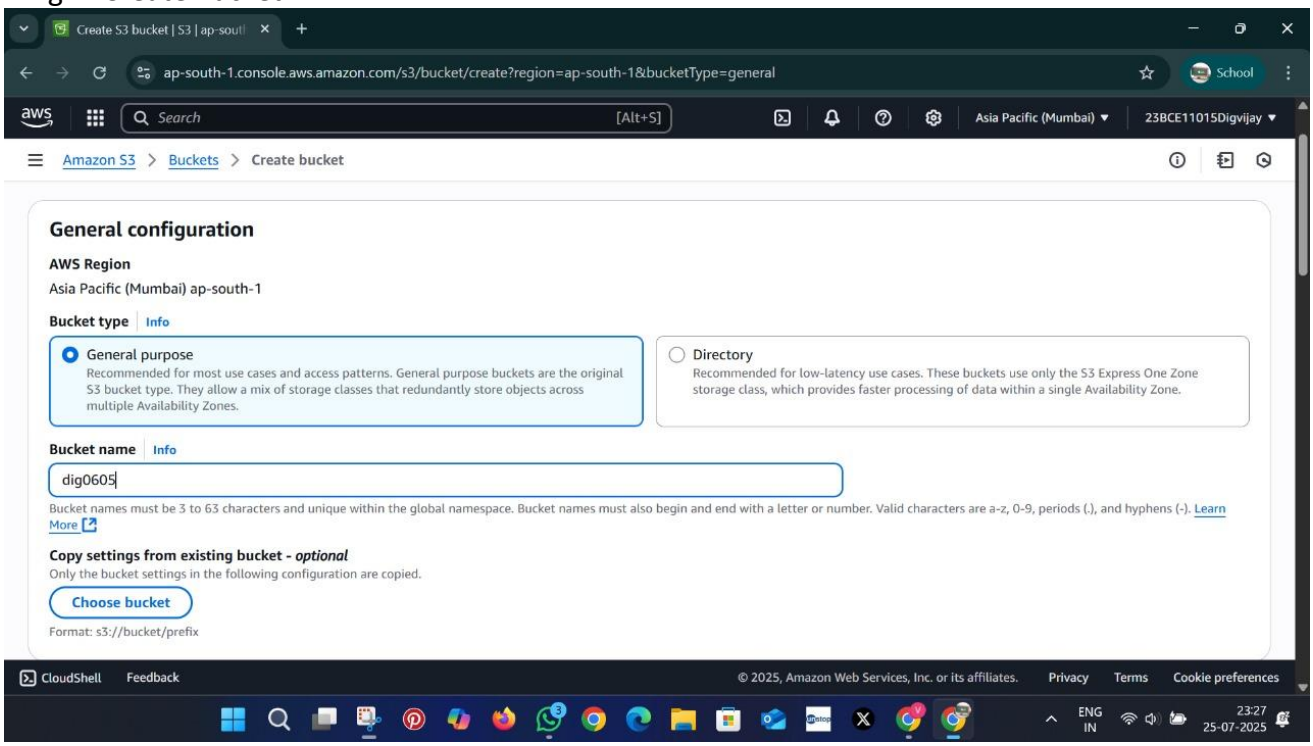


Fig 2: Provide Bucket Name

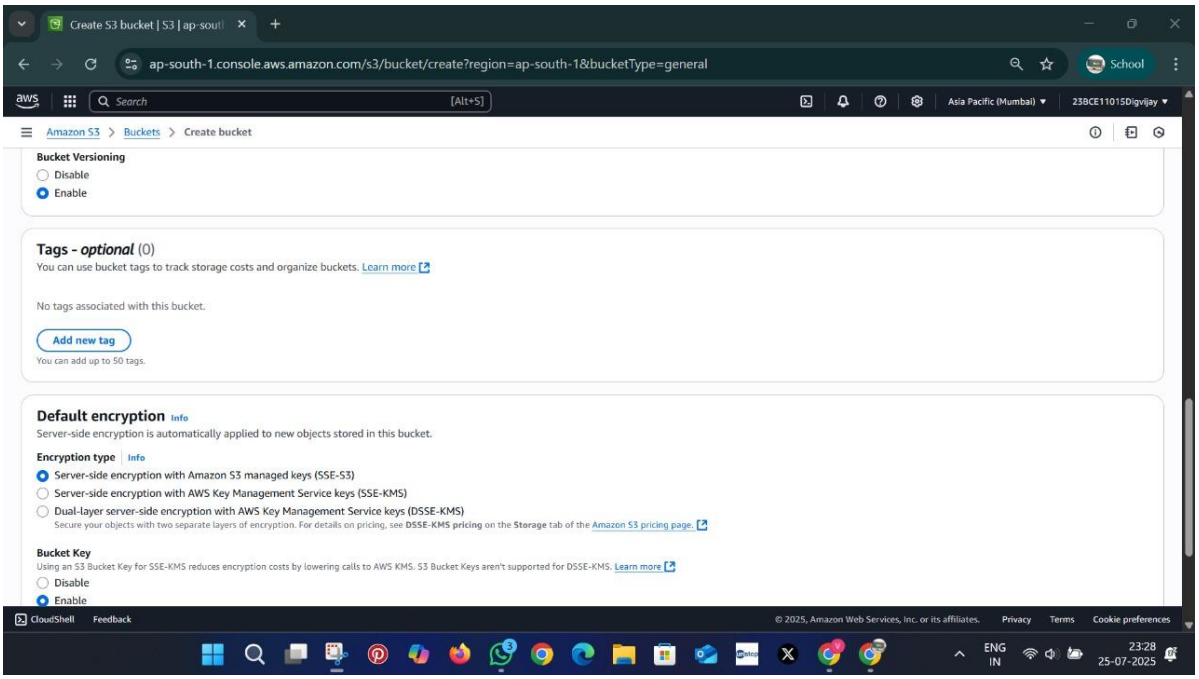


Fig 3: Enable Versioning

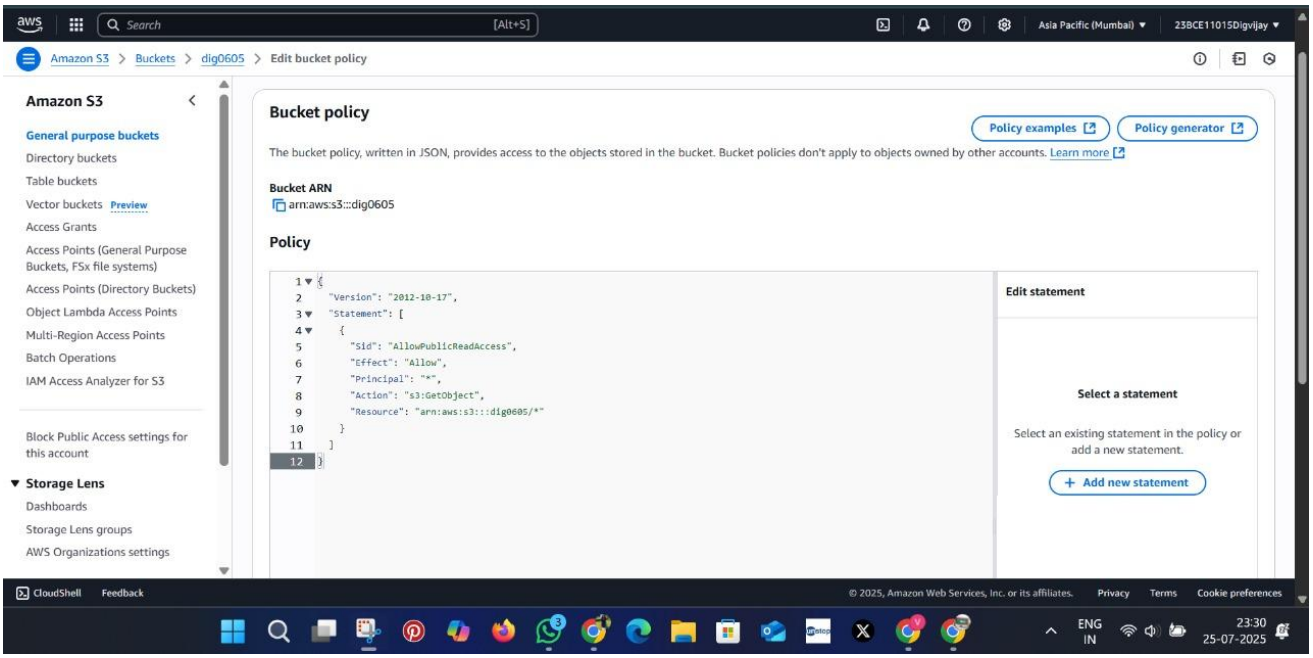


Fig 4: Update Bucket Policies

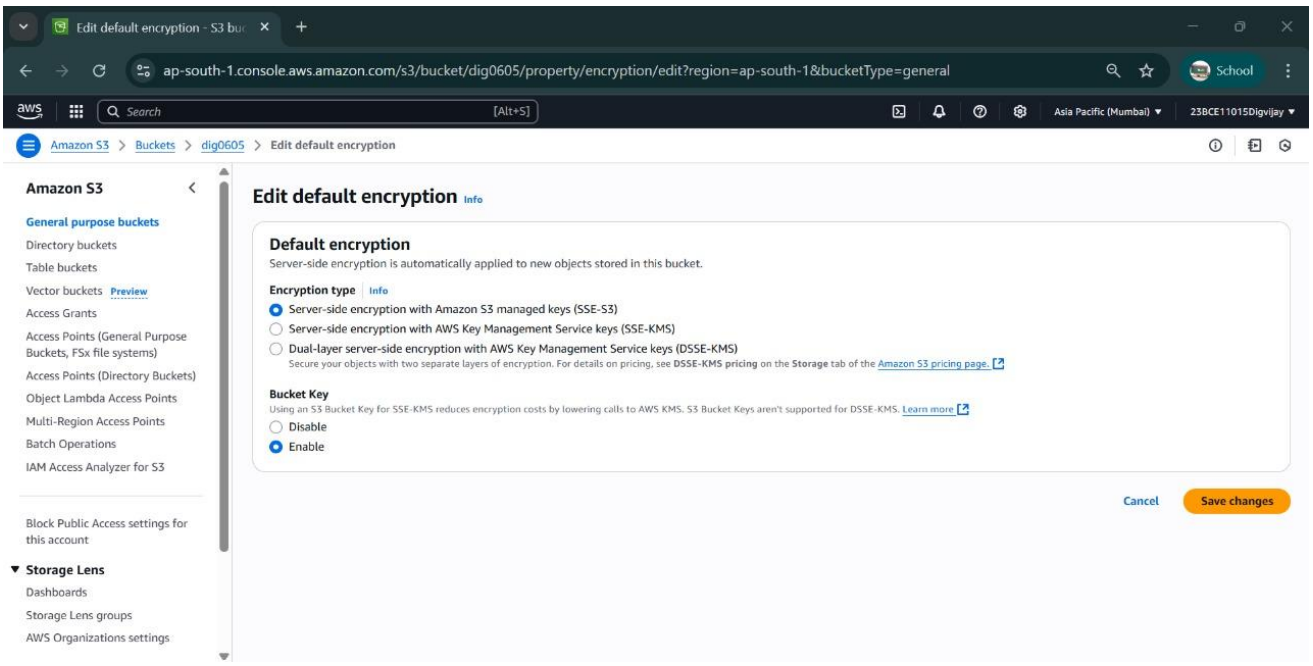


Fig 5: Enable File Encryption

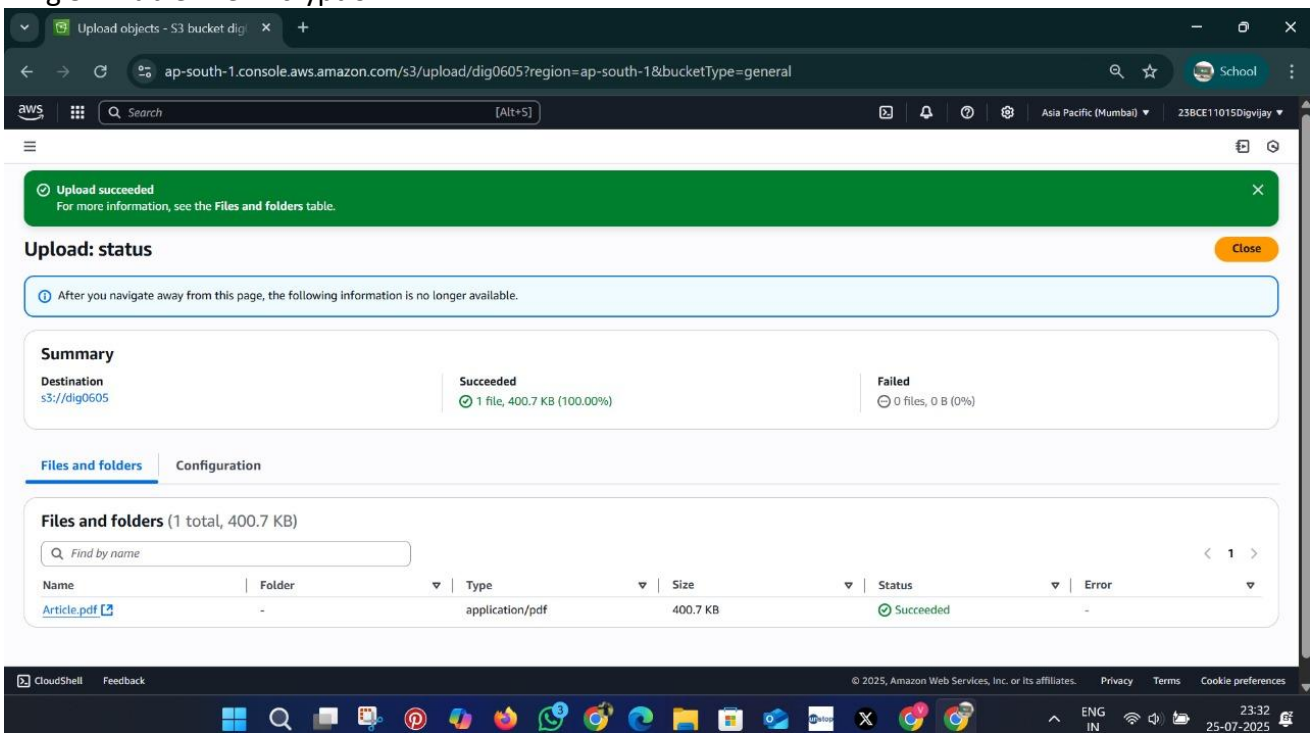


Fig 6: Upload File via AWS Management Console

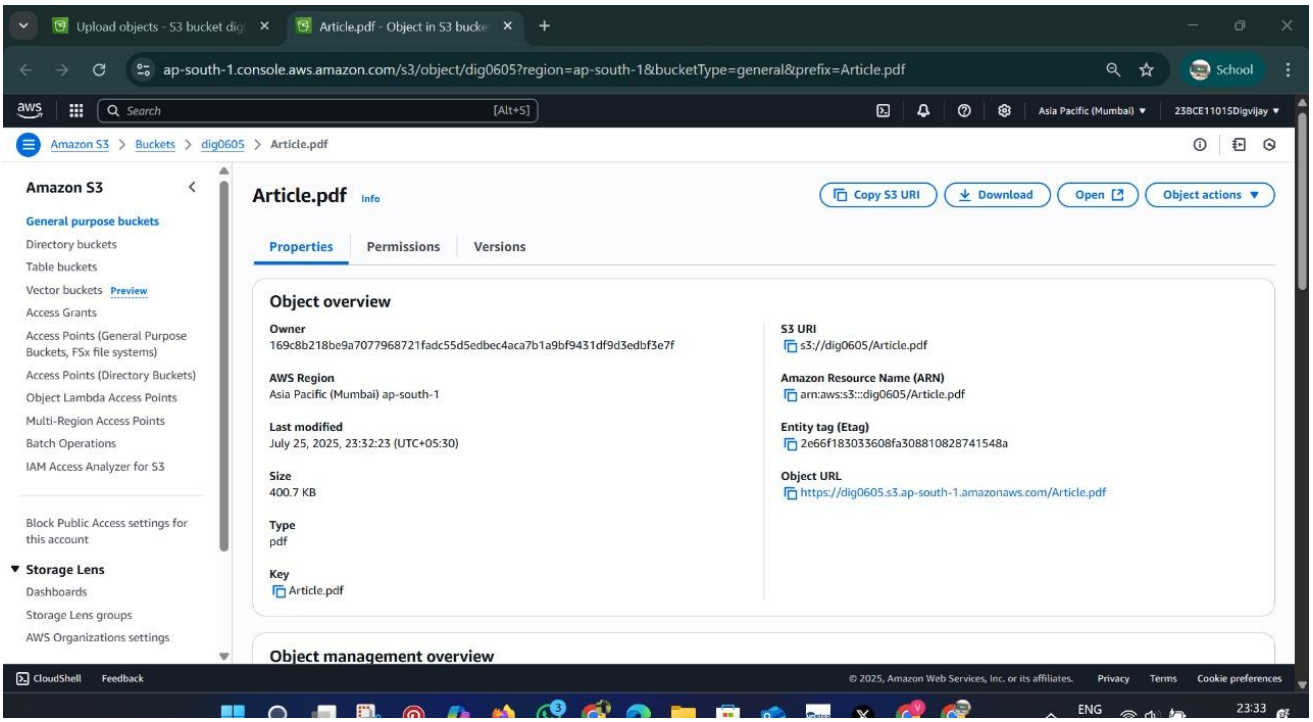

Upload objects - S3 bucket dig
Article.pdf - Object in S3 bucke
ap-south-1.console.aws.amazon.com/s3/object/dig0605?region=ap-south-1&bucketType=general&prefix=Article.pdf
School
Search
[Alt+S]
Asia Pacific (Mumbai)
23BCE11015Digvijay
Amazon S3 > Buckets > dig0605 > Article.pdf
Amazon S3
General purpose buckets
Directory buckets
Table buckets
Vector buckets Preview
Access Grants
Access Points (General Purpose Buckets, FSx file systems)
Access Points (Directory Buckets)
Object Lambda Access Points
Multi-Region Access Points
Batch Operations
IAM Access Analyzer for S3
Block Public Access settings for this account
Storage Lens
Dashboards
Storage Lens groups
AWS Organizations settings
Article.pdf Info
Copy S3 URI
Download
Open
Object actions
Properties
Permissions
Versions
Object overview
Owner
169c8b218be9a7077968721fadc55d5edbec4aca7b1a9bf9431df9d3edbf3e7f
AWS Region
Asia Pacific (Mumbai) ap-south-1
Last modified
July 25, 2025, 23:32:23 (UTC+05:30)
Size
400.7 KB
Type
pdf
Key
Article.pdf
S3 URI
s3://dig0605/Article.pdf
Amazon Resource Name (ARN)
arn:aws:s3:::dig0605/Article.pdf
Entity tag (Etag)
2e66f183033608fa308810828741548a
Object URL
https://dig0605.s3.ap-south-1.amazonaws.com/Article.pdf
Object management overview
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
ENG
23:33

Fig 7: View Uploaded File

Output:

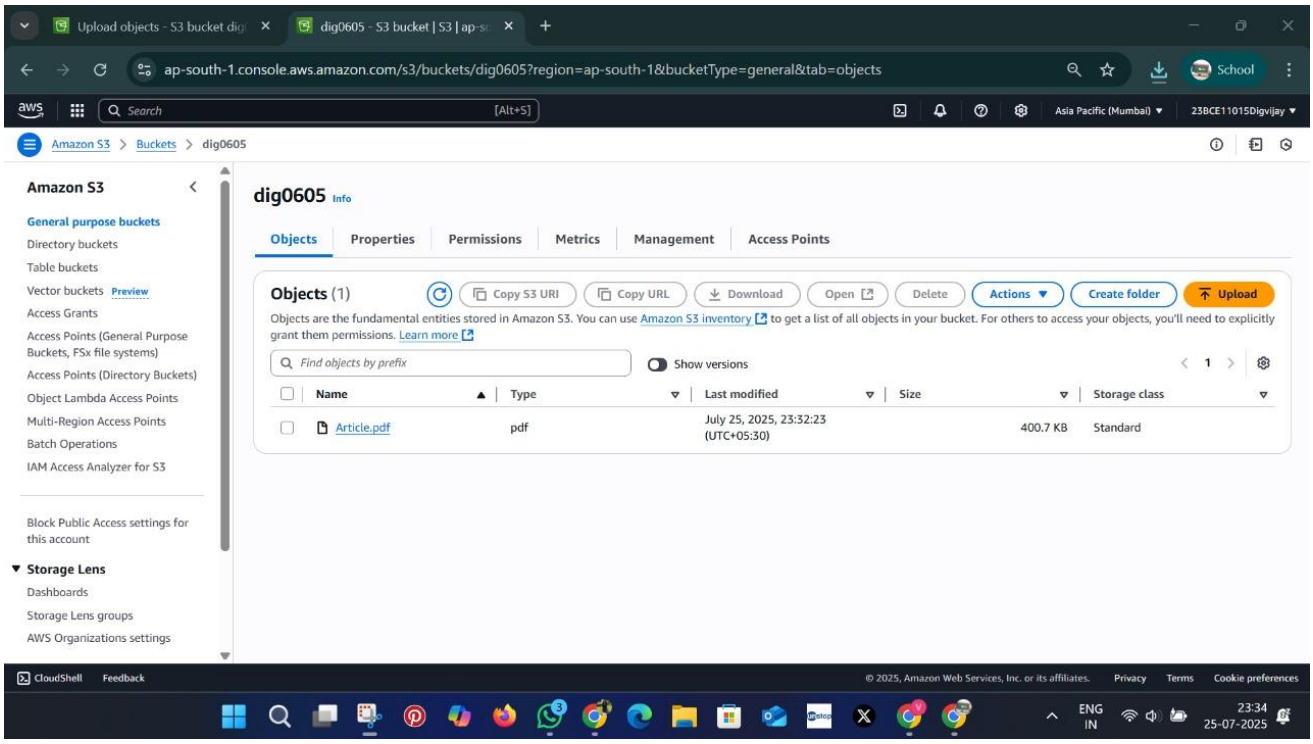


Fig 8: Uploaded File Successfully.

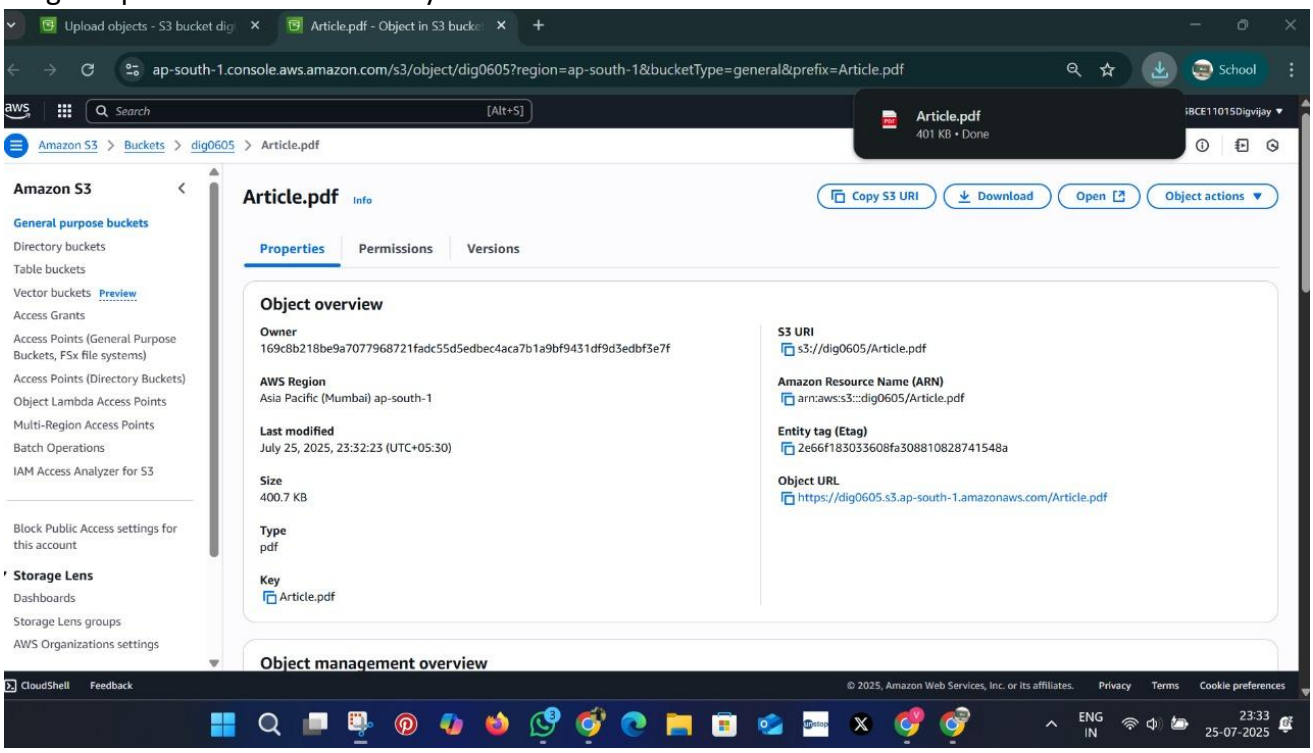


Fig 9: Successfully Downloaded.

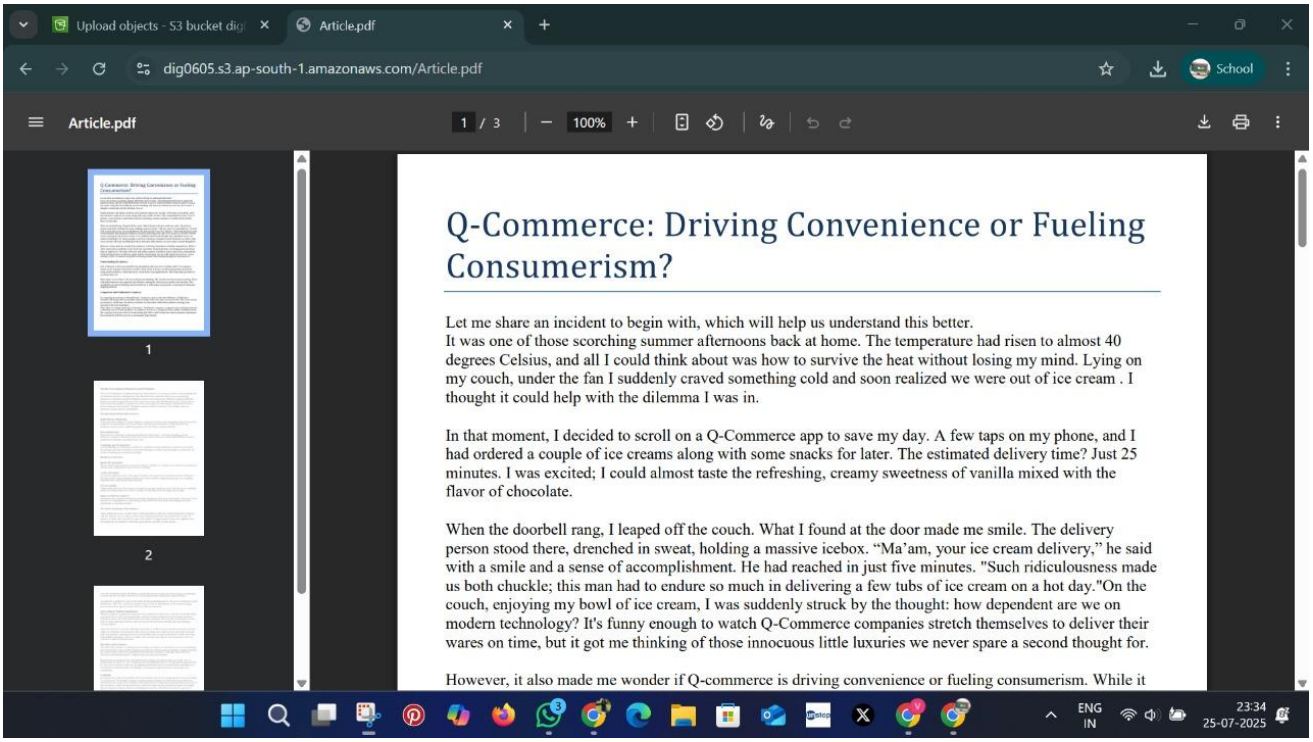


Fig 10: View contents of uploaded file.

Remarks:

The AWS Management Console was used for all bucket operations.

Conclusion:

The objectives mentioned above are achieved successfully.

Challenges:

1. Security Management: Configuring correct bucket policies and access controls can be complex and prone to misconfiguration.
2. Cost Optimization: Unmonitored storage growth and data transfer costs can lead to unexpectedly high bills.
3. Latency Issues: Retrieving large objects from S3 can result in latency, especially for real-time applications.

Hands-on Lab Manual (Experiment 3)

Create a Lambda function using Python or Node.js and Configure event triggers and test the Lambda function.Exercise (Objective) 3: To create a Lambda function using Python or Node.js and Configure event triggers and test the Lambda function.

Prerequisites:
You should already be familiar with:
- Basic understanding of **cloud computing** concepts.
- Basic programming in **Python** or **Node.js**.

Objectives:
After completion of this lab student shall be able to:
- Explain the concept of AWS Lambda.
- Demonstrate the creation and deployment of a Lambda function.
- Implement and test Lambda functions triggered by events.

Underlying Concept / Theory:
AWS Lambda is a serverless compute service that lets you run code without provisioning or managing servers. You simply upload your code as a Lambda function and AWS handles everything needed to run and scale the execution.

Key concepts include:

- **Function creation**
- **Handler definition**
- **Trigger configuration**
- **Testing and monitoring**

Problem Statement:

Program 3: Create an AWS Lambda function using Python or Node.js, set up event triggers, and perform testing to ensure proper execution.

Algorithm:

1. Create the Lambda Function:

- **Access the AWS Lambda Console:**
- Sign in to the AWS Management Console and open the Lambda console.
- **Create Function:**
- Click "Create function."
- Select "Author from scratch."AWS Documentation
- Enter a function name (e.g., MyFunction).
- Choose the runtime:AWS Documentation
- For Python, select Python 3.x.
- Set up permissions by selecting or creating an execution role.
- Click "Create function."

2. Add Function Code:

- In the function's code editor:
- For Python, edit lambda_function.py.
- Ensure the handler is set correctly:
- For Python, use lambda_function.lambda_handler.
- Click "Deploy" to save changes.

3. Configure an Event Trigger (e.g., API Gateway):

- In the Lambda function's page, under "Function overview," click "Add trigger."
- Select "API Gateway."
- Choose "Create an API" and select "HTTP API."
- Set security to "Open."
- Click "Add."

4. Test the Function:

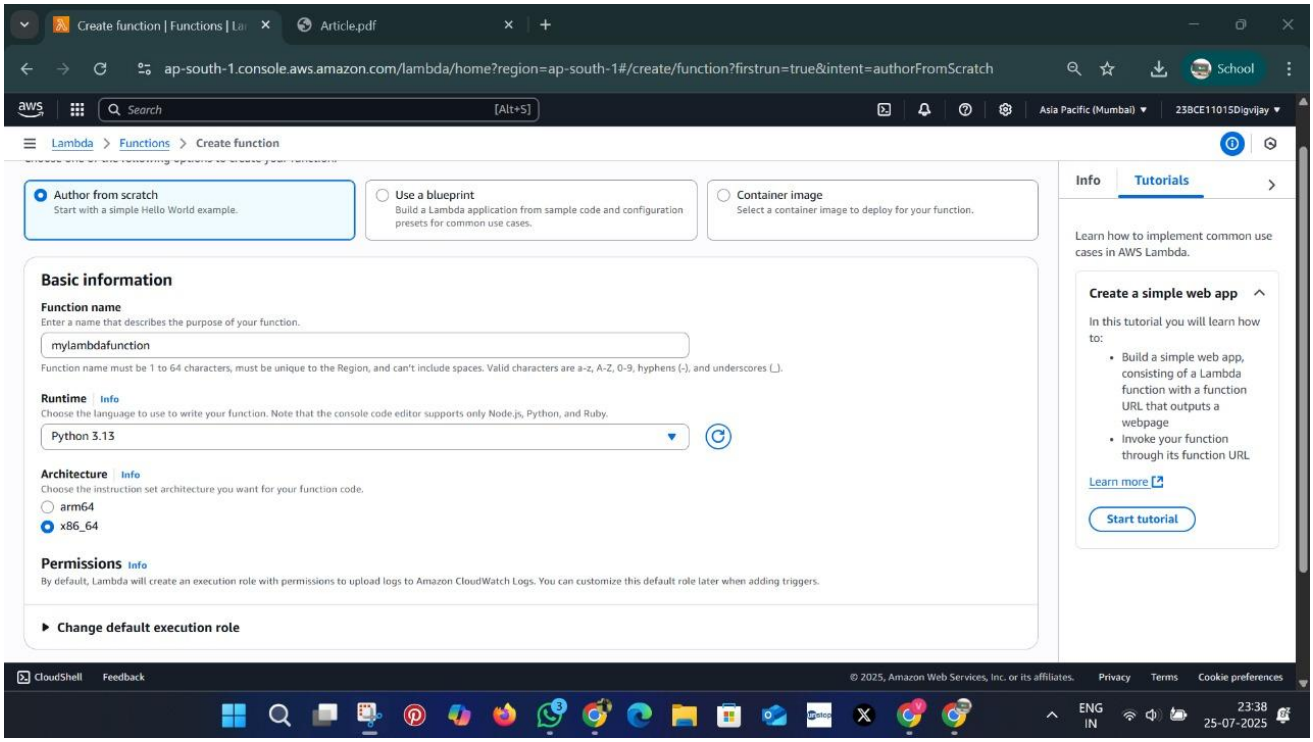
- **Direct Invocation:**
- In the Lambda console, click "Test."

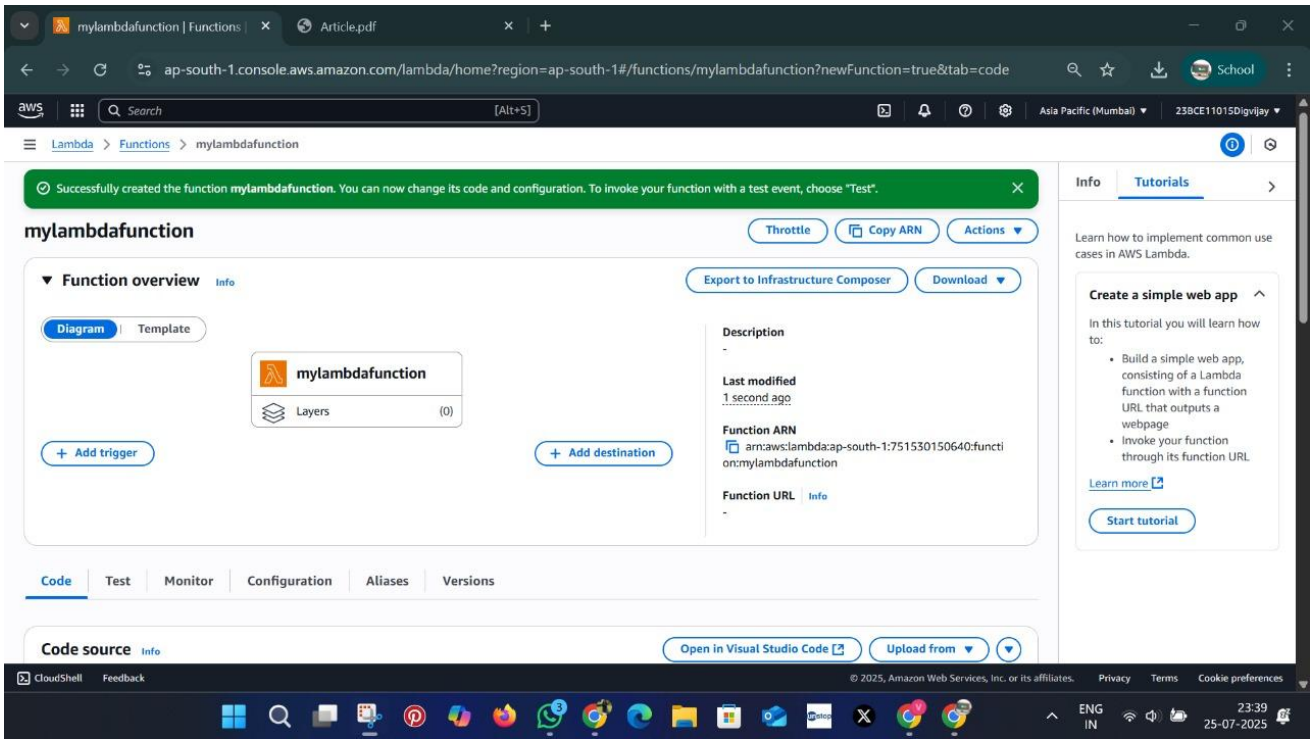
- Configure a test event with sample input data.
- Invoke the function and review the output.
- **Via API Gateway:**
- Use the provided API endpoint to send HTTP requests (e.g., using a browser or tools like cURL or Postman).
- Verify the function's response.

Coding:

```
import json
def lambda_handler(event, context):
    return {
        "statusCode": 200,
        "body": json.dumps("Hello from Lambda!")
    }
```

Execution Snapshots:




mylambdafunction | Functions
Article.pdf
ap-south-1.console.aws.amazon.com/lambda/home?region=ap-south-1#/functions/mylambdafunction?newFunction=true&tab=code
Lambda > Functions > mylambdafunction
Successfully created the function mylambdafunction. You can now change its code and configuration. To invoke your function with a test event, choose "Test".
mylambdafunction
Throttle
Copy ARN
Actions
Function overview
Export to Infrastructure Composer
Download
Diagram
Template
mylambdafunction
Layers (0)
Add trigger
Add destination
Description
Last modified
1 second ago
Function ARN
arn:aws:lambda:ap-south-1:751530150640:function:mylambdafunction
Function URL
Code
Test
Monitor
Configuration
Aliases
Versions
Code source
Open in Visual Studio Code
Upload from
Info
Tutorials
Learn how to implement common use cases in AWS Lambda.
Create a simple web app
In this tutorial you will learn how to:
Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
Invoke your function through its function URL
Learn more
Start tutorial
CloudShell
Feedback
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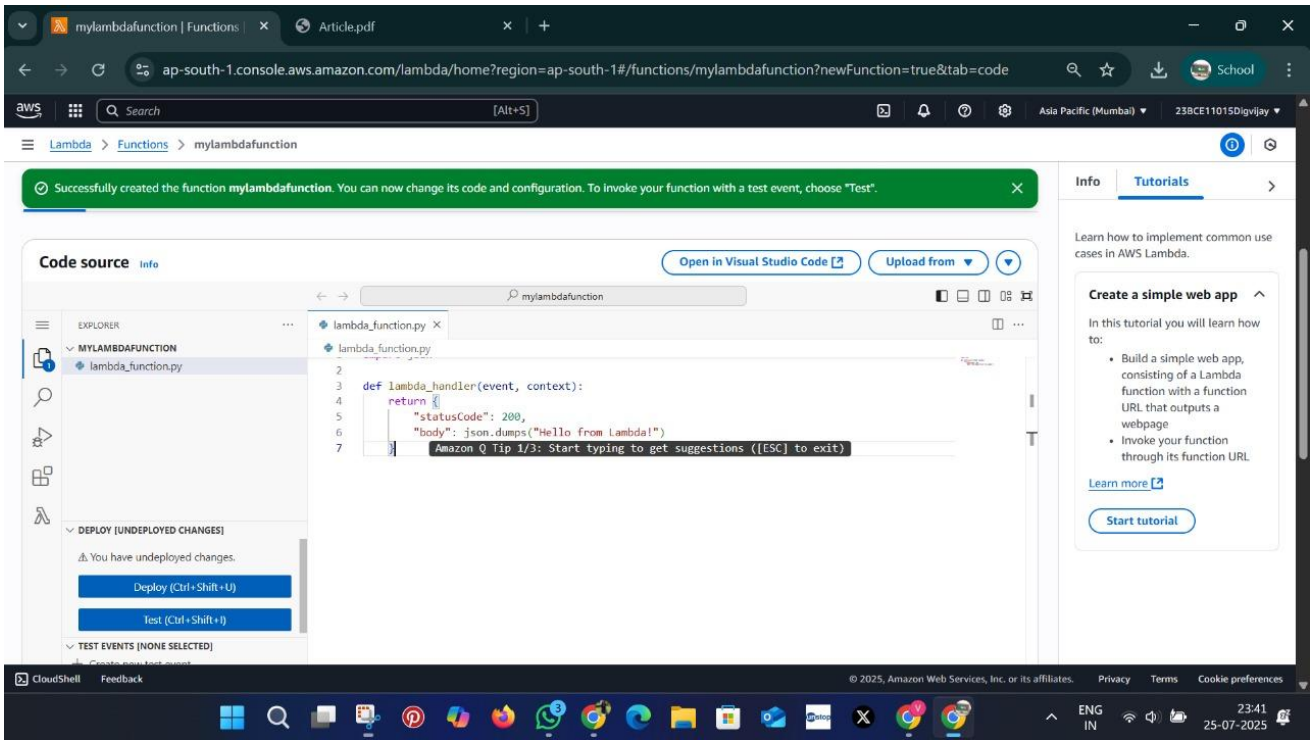
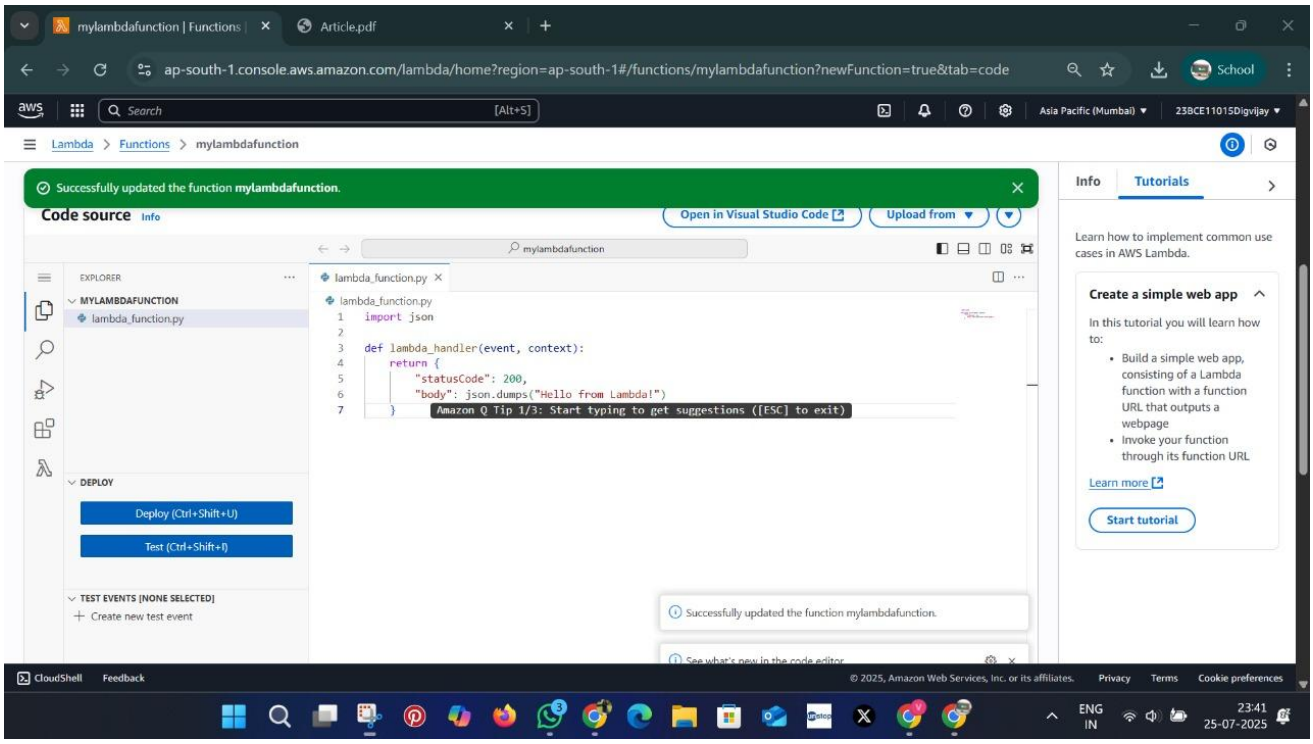
Cookie preferences


mylambdafunction | Functions
Article.pdf
ap-south-1.console.aws.amazon.com/lambda/home?region=ap-south-1#/functions/mylambdafunction?newFunction=true&tab=code
Lambda > Functions > mylambdafunction
Successfully created the function mylambdafunction. You can now change its code and configuration. To invoke your function with a test event, choose "Test".
Code source
Open in Visual Studio Code
Upload from
EXPLORER
MYLAMBDAFUNCTION
lambda_function.py
def lambda_handler(event, context):
return {
"statusCode": 200,
"body": json.dumps("Hello from Lambda!")
}
Amazon Q Tip 1/3: Start typing to get suggestions ([ESC] to exit)
DEPLOY [UNDEPLOYED CHANGES]
You have undeployed changes.
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
TEST EVENTS [NONE SELECTED]
Info
Tutorials
Learn how to implement common use cases in AWS Lambda.
Create a simple web app
In this tutorial you will learn how to:
Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
Invoke your function through its function URL
Learn more
Start tutorial
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


mylambdafunction | Functions
Article.pdf
ap-south-1.console.aws.amazon.com/lambda/home?region=ap-south-1#/functions/mylambdafunction?newFunction=true&tab=code
Lambda > Functions > mylambdafunction
Successfully updated the function mylambdafunction.
Code source Info
Open in Visual Studio Code
Upload from
EXPLORER
MYLAMBDAFUNCTION
lambda_function.py
import json
def lambda_handler(event, context):
return {
"statusCode": 200,
"body": json.dumps("Hello from Lambda!")
}
Amazon Q Tip 1/3: Start typing to get suggestions ([ESC] to exit)
DEPLOY
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
TEST EVENTS [NONE SELECTED]
Create new test event
Info
Tutorials
Learn how to implement common use cases in AWS Lambda.
Create a simple web app
In this tutorial you will learn how to:
Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
Invoke your function through its function URL
Learn more
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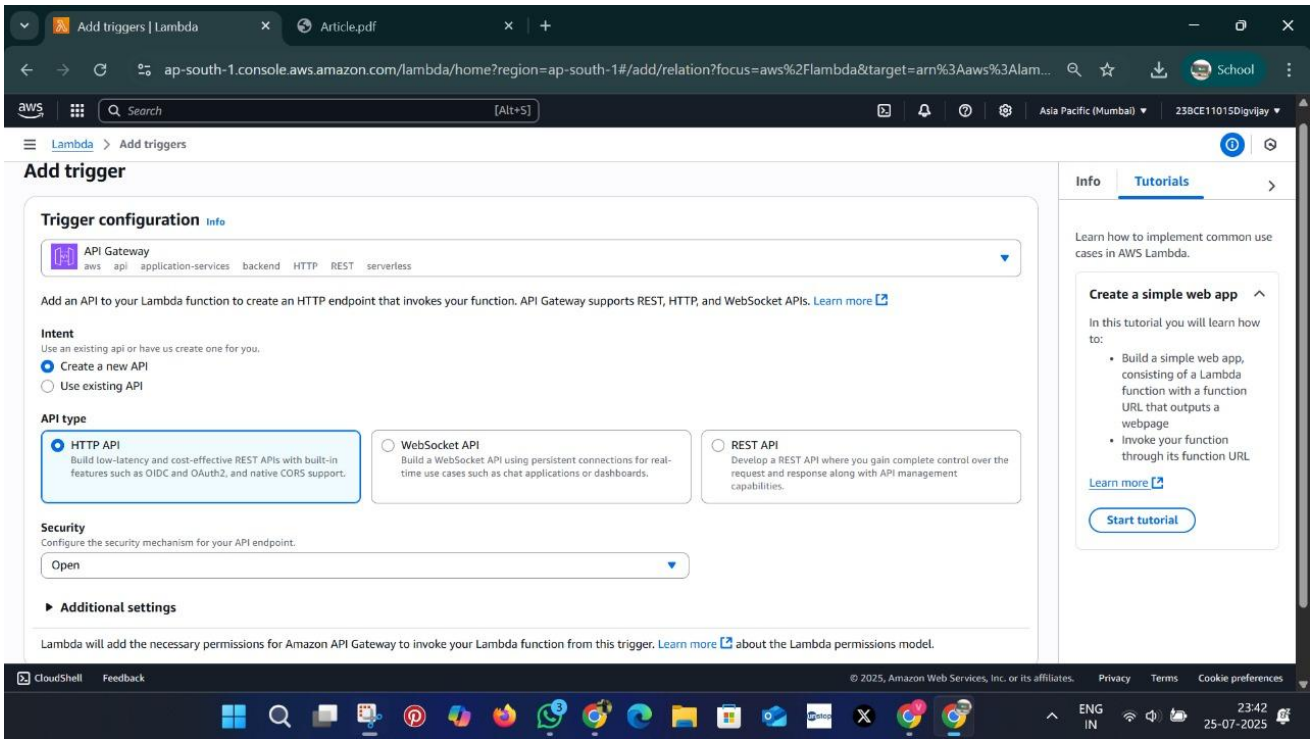
Add triggers | Lambda
Article.pdf
ap-south-1.console.aws.amazon.com/lambda/home?region=ap-south-1#/add/relation?focus=aws%2Flambda&target=arn%3Aaws%3Alam...
Lambda > Add triggers
Add trigger
Trigger configuration Info
API Gateway
aws api application-services backend HTTP REST serverless
Add an API to your Lambda function to create an HTTP endpoint that invokes your function. API Gateway supports REST, HTTP, and WebSocket APIs. Learn more
Intent
Use an existing api or have us create one for you.
Create a new API
Use existing API
API type
HTTP API
Build low-latency and cost-effective REST APIs with built-in features such as OIDC and OAuth2, and native CORS support.
WebSocket API
Build a WebSocket API using persistent connections for real-time use cases such as chat applications or dashboards.
REST API
Develop a REST API where you gain complete control over the request and response along with API management capabilities.
Security
Configure the security mechanism for your API endpoint.
Open
Additional settings
Lambda will add the necessary permissions for Amazon API Gateway to invoke your Lambda function from this trigger. Learn more about the Lambda permissions model.
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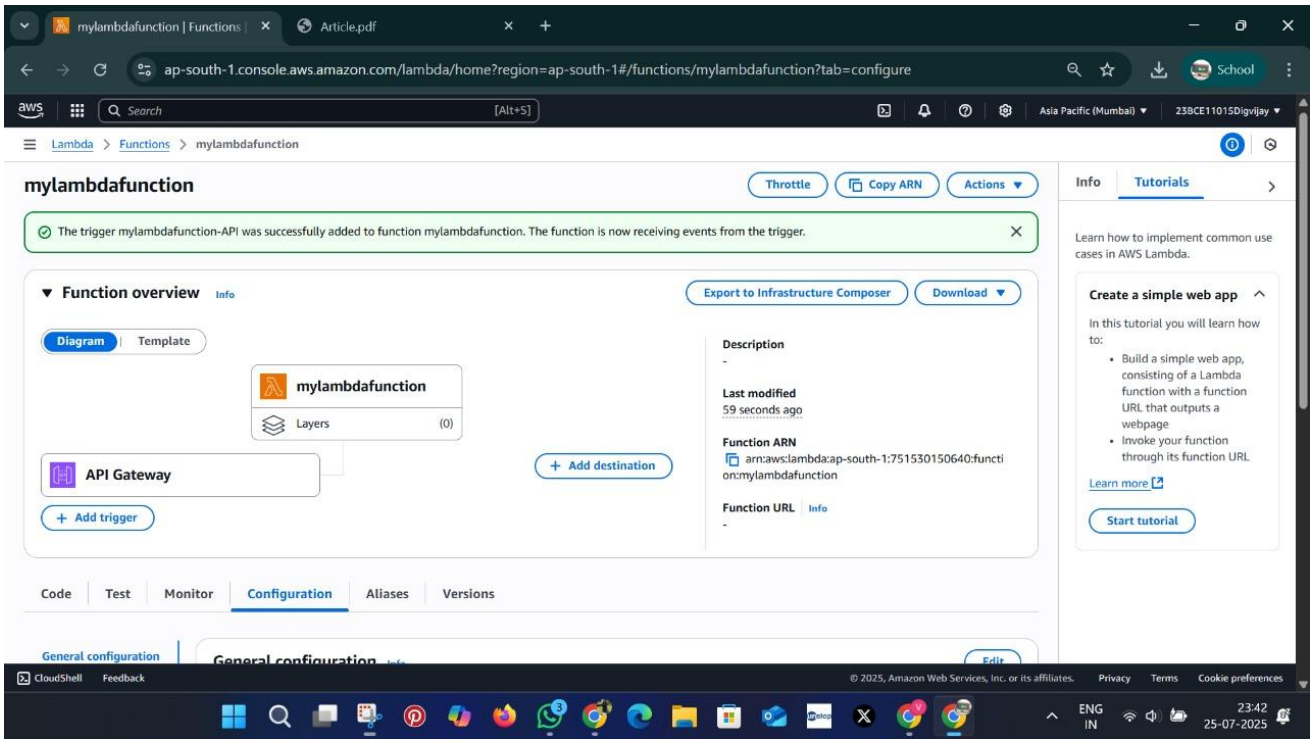
mylambdafunction | Functions
Article.pdf
ap-south-1.console.aws.amazon.com/lambda/home?region=ap-south-1#/functions/mylambdafunction?tab=configure
Lambda > Functions > mylambdafunction
mylambdafunction
Throttle
Copy ARN
Actions
The trigger mylambdafunction-API was successfully added to function mylambdafunction. The function is now receiving events from the trigger.
Function overview
Export to Infrastructure Composer
Download
Diagram
Template
mylambdafunction
Layers (0)
API Gateway
Add destination
Add trigger
Description
Last modified
59 seconds ago
Function ARN
arn:aws:lambda:ap-south-1:751530150640:function:mylambdafunction
Function URL
Code
Test
Monitor
Configuration
Aliases
Versions
General configuration
Info
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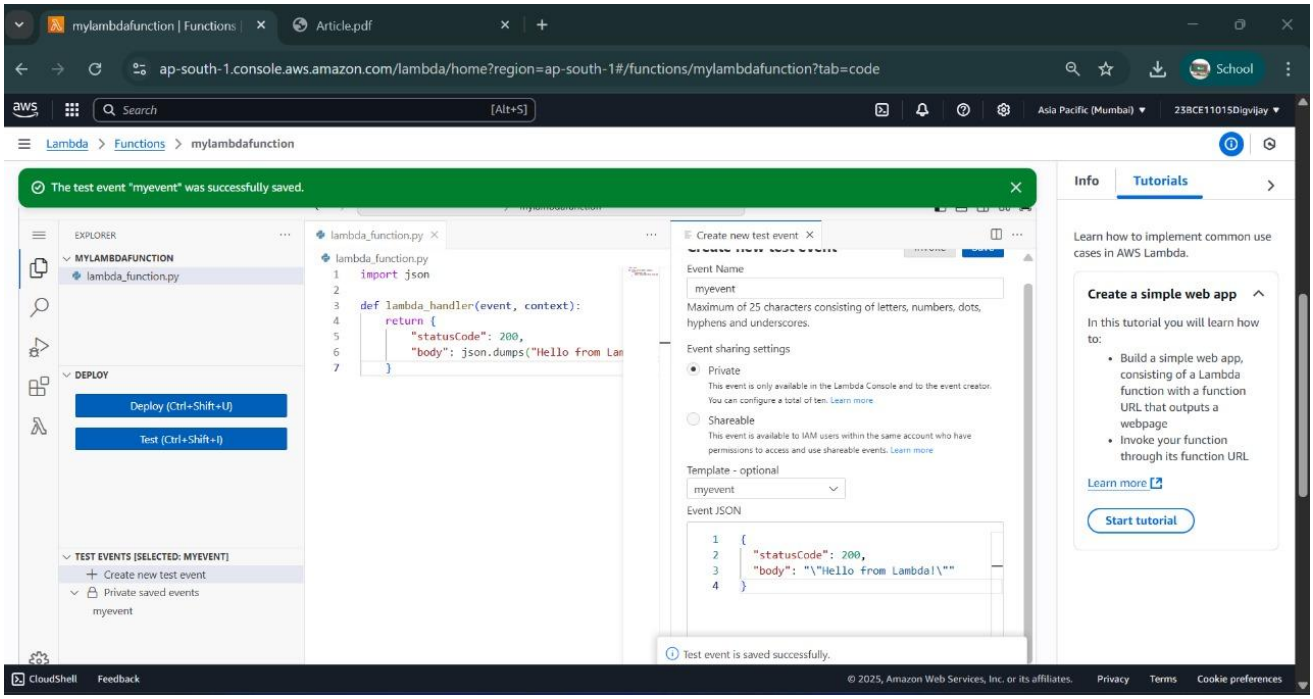
mylambdafunction | Functions
Article.pdf
ap-south-1.console.aws.amazon.com/lambda/home?region=ap-south-1#/functions/mylambdafunction?tab=code
Lambda > Functions > mylambdafunction
The test event "myevent" was successfully saved.
EXPLORER
MYLAMBDAFUNCTION
lambda_function.py
import json
def lambda_handler(event, context):
return {
"statusCode": 200,
"body": json.dumps("Hello from Lam
}
DEPLOY
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
TEST EVENTS [SELECTED: MYEVENT]
Create new test event
Private saved events
myevent
Create new test event
Event Name
myevent
Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.
Event sharing settings
Private
Shareable
Template - optional
myevent
Event JSON
{
"statusCode": 200,
"body": "\"Hello from Lambda!\""
}
Test event is saved successfully.
Info
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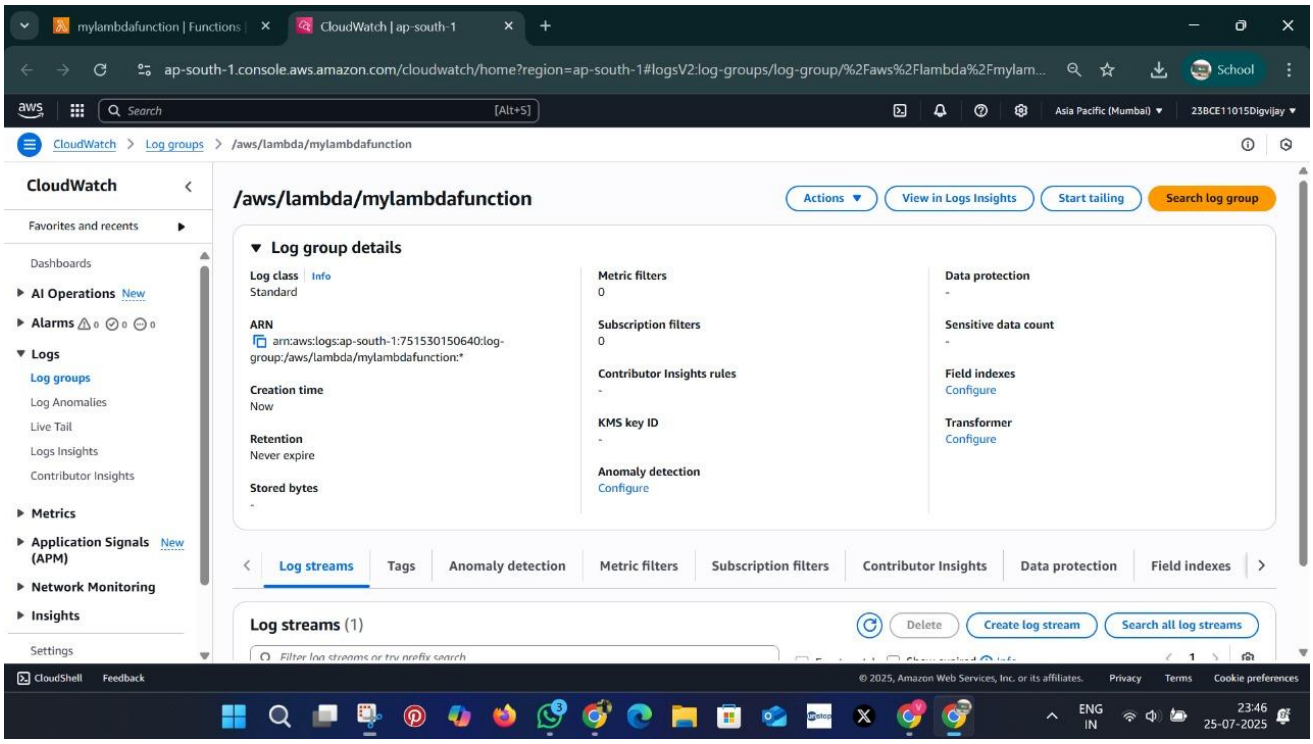
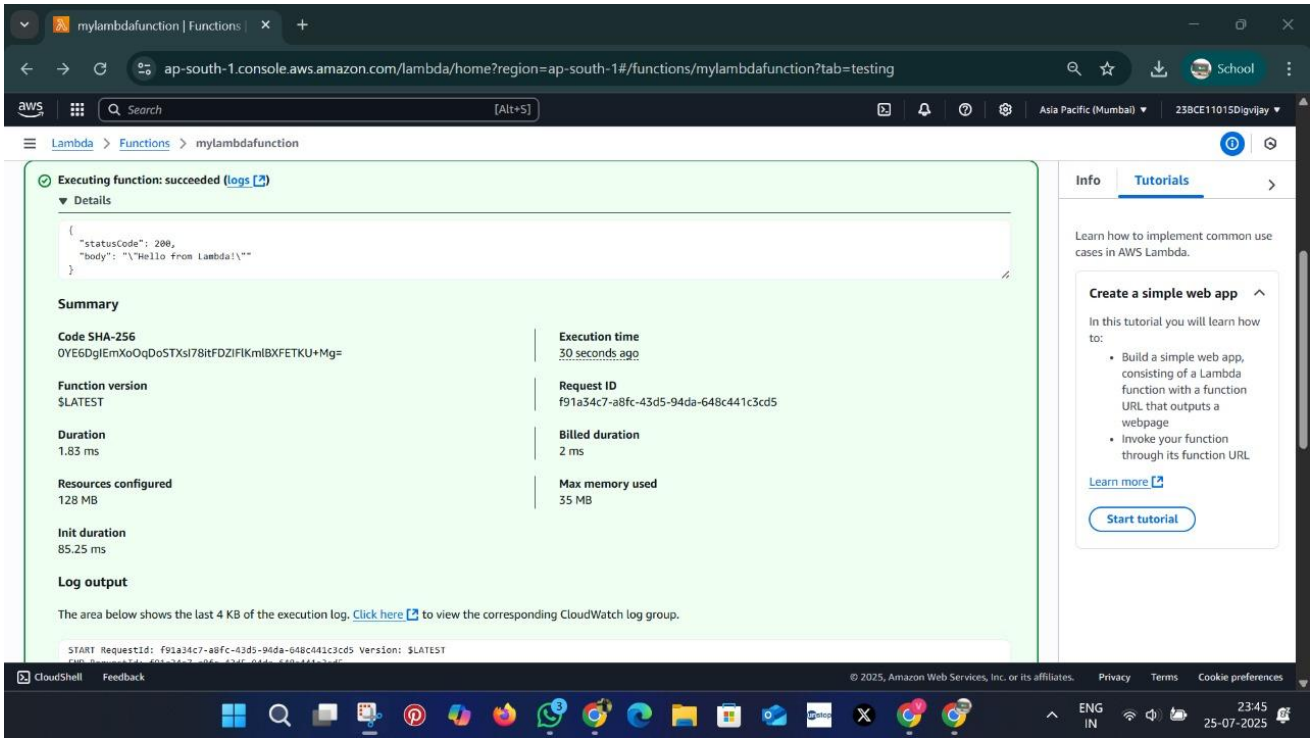

mylambdafunction | Functions
CloudWatch | ap-south-1
ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#logsV2:log-groups/log-group/%2Faws%2Flambda%2Fmylam...
CloudWatch > Log groups > /aws/lambda/mylambdafunction
/aws/lambda/mylambdafunction
Actions
View in Logs Insights
Start tailing
Search log group
Log group details
Log class Info
Standard
ARN
arn:aws:logs:ap-south-1:751530150640:log-group:/aws/lambda/mylambdafunction:*
Creation time
Now
Retention
Never expire
Stored bytes
-
Metric filters
0
Subscription filters
0
Contributor Insights rules
-
KMS key ID
-
Anomaly detection
Configure
Data protection
-
Sensitive data count
-
Field indexes
Configure
Transformer
Configure
Log streams
Tags
Anomaly detection
Metric filters
Subscription filters
Contributor Insights
Data protection
Field indexes
Log streams (1)
Delete
Create log stream
Search all log streams

OUTPUT:


mylambdafunction | Functions
ap-south-1.console.aws.amazon.com/lambda/home?region=ap-south-1#/functions/mylambdafunction?tab=testing
Lambda > Functions > mylambdafunction
Executing function: succeeded (logs)
Details
{
"statusCode": 200,
"body": "\"Hello from Lambda!\""
}
Summary
Code SHA-256
0YE6DgIEmXoOqDoSTXsI78itFDZIFlKmlBXFETKU+Mg=
Execution time
30 seconds ago
Function version
$LATEST
Request ID
f91a34c7-a8fc-43d5-94da-648c441c3cd5
Duration
1.83 ms
Billed duration
2 ms
Resources configured
128 MB
Max memory used
35 MB
Init duration
85.25 ms
Log output
The area below shows the last 4 KB of the execution log. Click here to view the corresponding CloudWatch log group.
START RequestId: f91a34c7-a8fc-43d5-94da-648c441c3cd5 Version: $LATEST
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